Plant Disease Detection and Classification

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Abstract

In this study, the problem of plant disease image classification was investigated using Deep Learning methods. During the research, image data preprocessing was performed, which transformed the initial dataset into a structured format suitable for model training, utilizing disease-only labeling. Both Convolutional Neural Networks (EfficientNet-B0, GoogLeNet, MobileNetV3-Large) and transformer architectures—specifically the CLIP (ViT-B/32) and DINOv2-Large models—were applied. For the large-scale transformer models, the LoRA (Low-Rank Adaptation) method was utilized as a parameter-efficient fine-tuning strategy. The model training and evaluation results indicate that the CNN models provide roughly comparable accuracy (73–78%), but exhibit certain signs of overfitting caused by data imbalance. CLIP, by training only about 0.21% of its parameters, achieved an accuracy equivalent to the CNNs, demonstrating the powerful generalization capability of vision-language pre-training. Meanwhile, the DINOv2 model yielded the best results, achieving approximately 90% accuracy on the test data and demonstrating significantly better generalization ability. The obtained findings show that self-supervised transformer models serve as a highly effective solution for complex image classification tasks, particularly in cases where there is data imbalance and high inter-class visual similarity.

1. Introduction

In image classification tasks, deep neural networks are widely used due to their ability to automatically learn high-level feature representations. Convolutional Neural Networks (CNNs) are effective in modeling local visual structures thanks to their inductive bias, which focuses on local features, edges, and textures. Meanwhile, transformer models are based on the self-attention mechanism and allow for the capture of global dependencies present in an image without the strict constraints of spatial proximity. In recent years, vision-language and self-supervised pre-trained transformers, such as CLIP and DINOv2, have attracted particularly great attention, acquiring high-quality, generalizable representations by being trained on massive amounts of data. Full fine-tuning of these models requires significant computational resources; however, the application of the LoRA (Low-Rank Adaptation) method enables the efficient adaptation of large models to new tasks by training only a small fraction of the parameters. Modern image datasets often possess complex structures, class imbalance, and high intra-class variance, which can negatively impact the training and generalization capabilities of models. Therefore, data preprocessing and the selection of the appropriate model are key factors in ensuring high performance. This work aims to investigate the effectiveness of various deep learning architectures for the problem of multi-class image classification. To this end, data preprocessing was carried out, resulting in a cleaned and structured dataset that ensures the absence of data leakage and the complete representation of all classes. Subsequently, five different architectures were applied: EfficientNet-B0, GoogLeNet, MobileNetV3-Large, CLIP (ViT-B/32 + LoRA), and DINOv2-Large (+ LoRA). The primary objective of the study is to provide a comparative analysis of these different rchitectures based on accuracy, F1-score, generalization ability, and computational efficiency, particularly under conditions of complex and imbalanced data.

2. Data Preprocessing

In image classification tasks, data preprocessing is just as critical as the model architecture itself. Particularly in plant disease recognition, a single disease can manifest across various plant species, while original datasets frequently feature complex folder structures, imbalanced class distributions, varying image dimensions, and duplicate samples. Consequently, without a well-defined preprocessing pipeline, a model may learn data source biases such as the plant species, background characteristics, or even duplicated images across different data splits rather than the actual visual characteristics of the disease. This chapter addresses precisely these challenges.

2.1 Formulation of Data Preprocessing Tasks

The primary objective was to transform the initial, partially heterogeneous dataset of plant disease images into a structured format suitable for subsequent machine learning, specifically for the training of CNN and ViT-type models. This objective entails the following tasks: (1) inspecting the structure of the initial folders and classes, (2) merging plant-specific labels into disease-only labels, (3) ensuring that all target classes are represented in the validation set, (4) standardizing the dimensions of all images without severe distortion of their aspect ratios, (5) removing duplicate samples, and (6) constructing a final, leakage-free train/validation/test split.

2.2 Analysis of the Initial Dataset

In the initial stage, an analysis of the raw dataset was conducted. The results revealed that the training split contained 82 original classes and 7,783 images, while the validation split contained 71 classes and 390 images. However, when only the disease names were extracted from the folder titles, it was found that 40 disease-only classes could be formed in the training set, whereas only 39 could be formed in the validation set. This discrepancy indicated that the initial validation split was incomplete, as one disease class was missing.

Table 1. Summary of the initial dataset structure.

	Original Classes	Disease-only Classes	Number of Images
Train	82	40	7,783
Validation	71	39	390

2.3 Class Aggregation and Disease-Level Identification

During data preprocessing, the automatic extraction of disease names from folder titles was performed, transforming the *plant+disease* format into *disease-only* labels. This step is methodologically significant, as it shifts the classification problem from identifying "which plant hosts the disease" to determining "what the disease is." Consequently, identical diseases occurring on different plants—such as *powdery mildew* or *black rot*—are consolidated into a single, common class. This increases the semantic density of the data and enables the model to focus on the visual features of the disease itself, rather than on the variations of the host plant species.

2.4 Completion and Balancing of the Validation Set

Following the initial analysis, it was observed that one disease-only class, specifically pepper_powdery_mildew, was missing from the validation set. This issue was resolved by transferring 10 images from the training split to the validation split. As a result, the cleaned validation set now encompasses all 40 target classes. This step is crucial; without it, the model validation phase could not comprehensively evaluate performance across all categories, rendering metrics such as recall and precision meaningless for the unrepresented classes.

2.5 Manual Cleaning Results and Intermediate Statistics

The clean dataset, generated following the initial data cleaning and preparation, comprises 8,366 training and 428 validation images, yielding a total of 8,794 samples. These figures exceed the initial counts of 7,773 and 400, respectively, indicating that additional steps were implemented during the data processing phase. The increase in the volume of data is attributed to the presence of images in the raw dataset that contained multiple sub-images (subplots). During preprocessing, these composite images were divided into separate, independent samples, resulting in a significant increase in the total number of instances. This approach facilitated the creation of a more granular and informative dataset, which is highly suitable for training machine learning models. This is a critical observation, as the subsequent analysis of results must be based on the final, updated state of the preprocessed data, rather than the initial raw data volume.

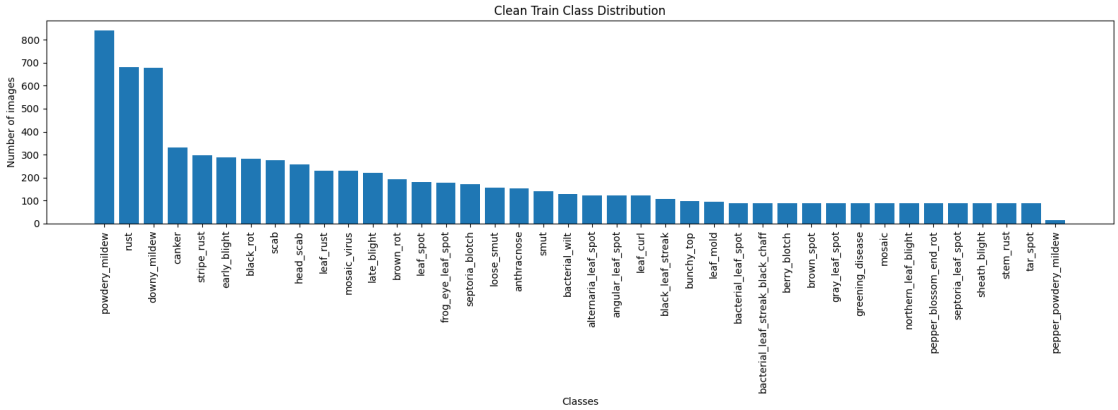

Figure 1. Class distribution graph of the clean training set. The graph demonstrates a noticeable imbalance among the classes.

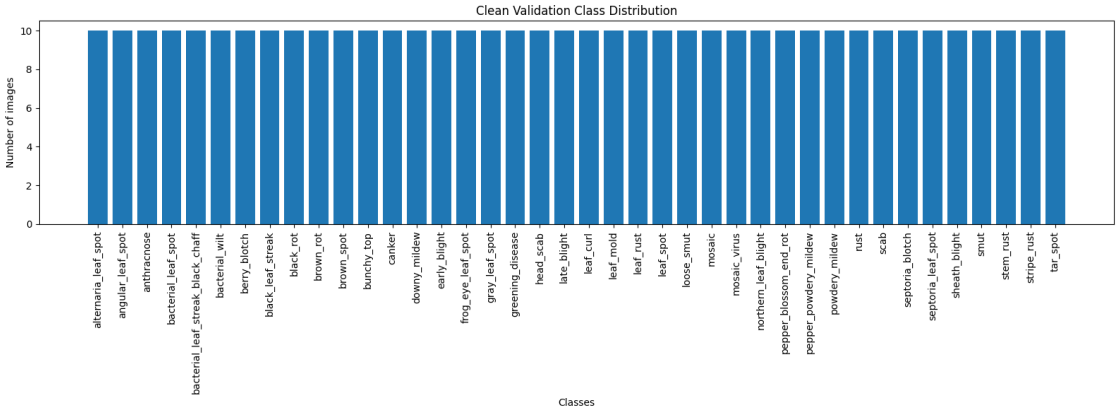

Figure 2. Class distribution of the clean validation set. While all 40 classes are represented, the sample counts are not yet uniform.

2.6 Image Size Analysis

An image size analysis was conducted on the cleaned dataset. A total of 8,794 images were examined, and their fundamental dimensional statistics were calculated. The results indicate a median width of 640 pixels and a median height of 501 pixels, demonstrating significant dimensional variance across the dataset. Such variation necessitates the standardization of image dimensions to ensure the stability and efficiency of the model training process. To achieve this, an aspect ratio-preserving approach was implemented, utilizing padding where necessary to conform the images to the target dimensions. This method prevents severe visual distortion and minimizes the risk of geometrically deforming the leaves and the diseased areas.

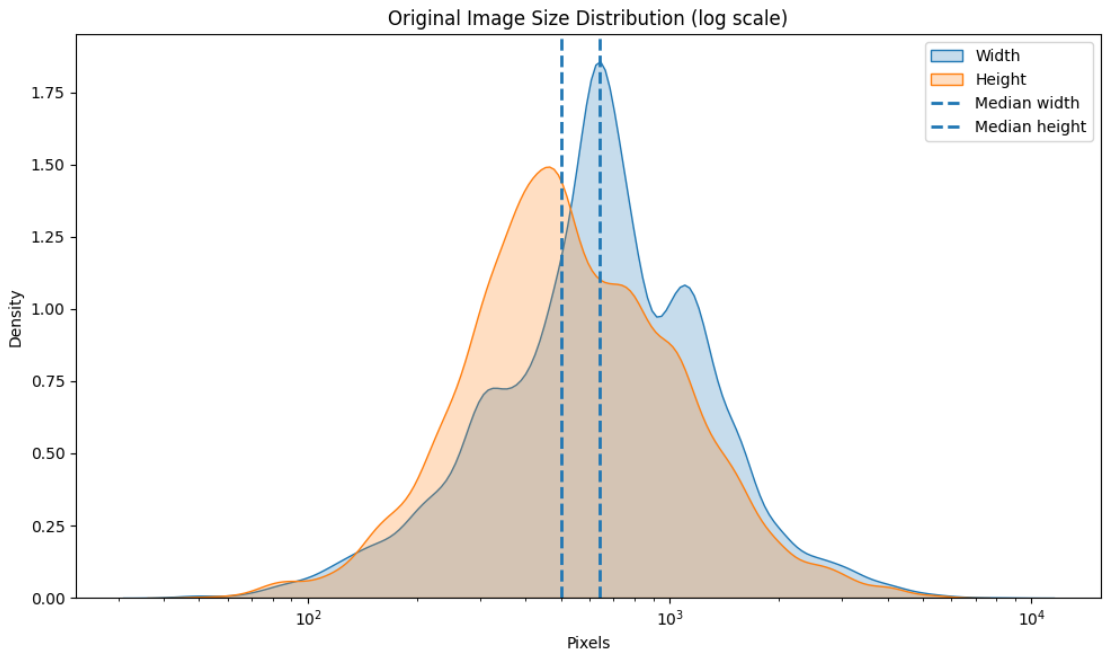

Figure 3. Density plot of the image size distribution. This justifies the necessity for image resizing.

2.7 Resizing to 224×224

During the resizing phase, all images were transformed to a resolution of 224×224 pixels utilizing an aspect ratio-preserving padding method. The 224×224 dimension is a widely adopted standard input size for numerous pre-trained Convolutional Neural Networks (CNN) and Vision Transformer (ViT) models. Thus, the implemented preprocessing pipeline not only ensures data normalization but also renders the dataset directly compatible with transfer learning scenarios. Following the resizing phase, a total of 8,794 images were processed, and all 40 classes were successfully preserved within the resulting dataset across both the training and validation splits.

2.8 Duplicate Removal and Final Data Split

The next crucial phase is duplicate control within the data. For this purpose, the MD5 hashes of the images were calculated, based on which duplicate samples were identified and removed. Furthermore, cross-label duplicates were also detected—that is, images with identical file content labeled under different classes. As a result, a dataset comprising 8,572 unique images was formed; 222 duplicate samples were removed, and 25 cross-label duplicates were isolated for further analysis. Based on the resulting cleaned data, the final

train_ready_data set was constructed, consisting of 6,000 training, 1,288 validation, and 1,284 testing images. Special attention was also given to verifying the absence of data leakage. Through hash comparisons, it was confirmed that there are no intersections among the train, validation, and test sets, meaning the leakage rate for all subset pairs is exactly 0. This guarantees the objectivity of the model evaluation and the reliability of the results.

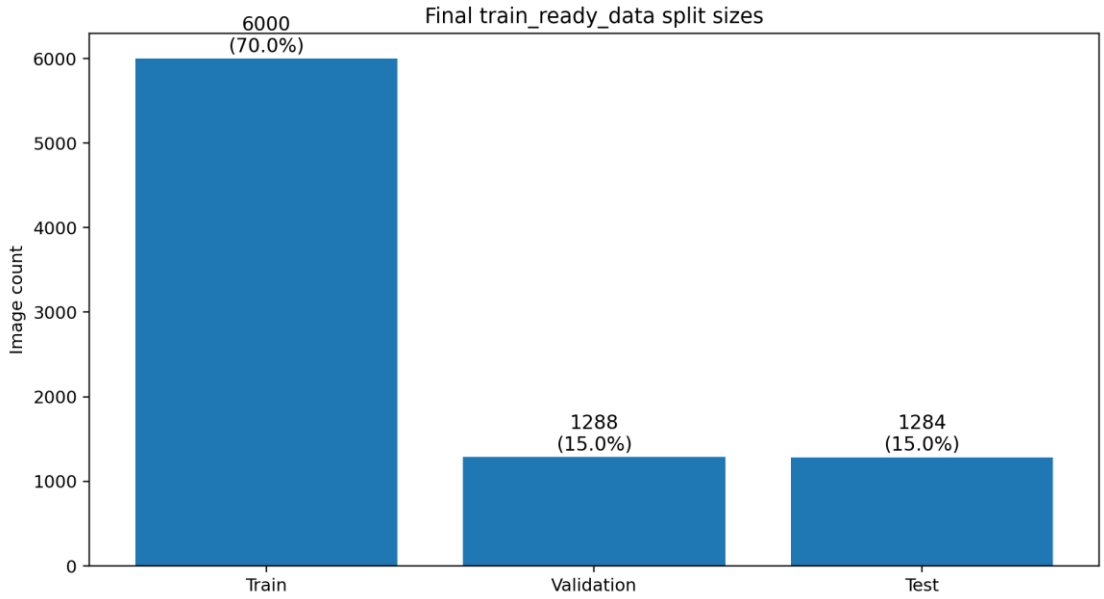

Figure 4. Volumes and percentage ratios of the final train/validation/test splits.

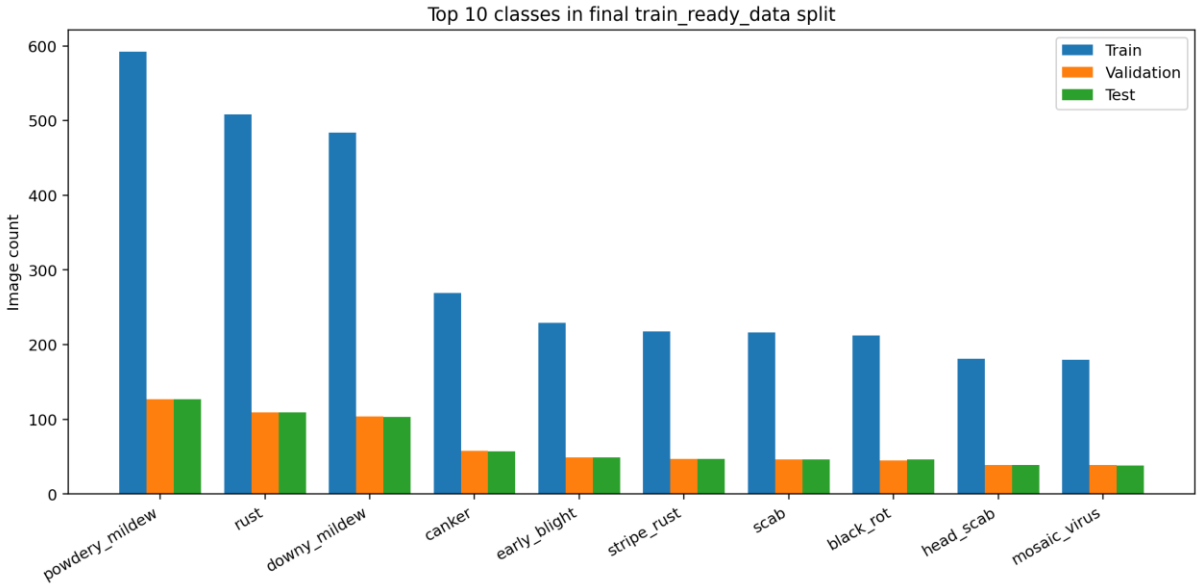

Figure 5. Distribution of the top 10 largest classes in the final `train_ready_data` dataset across the train, validation, and test splits.

2.9 Analysis of Data Preprocessing Results

The primary outcome of the analyzed preprocessing pipeline is that it transforms the initial, mixed-structure botanical image dataset into a structured, relatively clean, and machine learning-ready dataset. The resulting 40 disease-only classes possess higher semantic clarity than the original plant-specific labels, as the classification task is now directly aimed at disease recognition. Furthermore, the completion of the validation set ensures that all classes are represented during the evaluation phase, preventing situations where the model's performance on certain classes cannot be assessed. Analysis of the class distribution reveals the presence of class imbalance within the dataset. Specifically, the powdery mildew, rust, and downy mildew classes significantly predominate over several rarer classes. At the same time, it should be noted that the existence of this imbalance is not a flaw in the data preprocessing, but rather a characteristic feature of the dataset that has been properly identified and documented. From a methodological standpoint, the duplicate removal and data leakage control phase holds particular importance. If identical images were to appear simultaneously in the training and validation or test sets, the model's evaluations would be artificially inflated and unreliable. The zero-intersection results obtained through hash comparisons confirm that there is no overlap between the subsets, and the final split ensures conditions for objective evaluation. This factor is especially critical in scientific research projects, where preprocessing errors can significantly compromise the credibility of the results and even overshadow the impact of the model's architectural limitations.

3. MODEL SELECTION AND ANALYSIS OF RESULTS

The data preprocessing phase constitutes one of the fundamental components of a machine learning system, as properly processed data directly impacts the model's performance and efficiency. However, high-quality data preparation alone is not sufficient to guarantee superior results. The subsequent critical phase is model selection, which determines how the features extracted from the data will be utilized to effectively solve the classification problem. This chapter presents the rationale behind the selection of the applied models, their architectural characteristics, and a comparative analysis of the results obtained during the training process.

3.1 Rationale for Model Selection

In this study, five distinct architectures were selected to address the plant disease classification problem, representing both traditional Convolutional Neural Networks (CNN) and modern Transformer-based approaches: EfficientNet-B0, GoogLeNet, MobileNetV3-Large, CLIP ViT-B/32, and DINOv2-Large. This selection is driven by the fact that these models embody varying levels of complexity and fundamentally distinct approaches to image analysis. Convolutional networks are highly effective in extracting local features (such as edges, textures, and shapes), whereas transformer models, leveraging the self-attention mechanism, are capable of modeling global contextual dependencies within an image. Evaluating this diverse set of methodologies enables a comprehensive assessment of how effectively different architectural paradigms perform on this specific task.

3.2 Application of Convolutional Neural Networks (CNNs)

Within the framework of the automated plant disease classification problem, three distinct CNN architectures were investigated and evaluated: EfficientNet-B0, GoogLeNet, and MobileNetV3-Large. The selection of these specific models was driven by the necessity to establish an effective balance between achieving high classification accuracy and ensuring the optimal consumption of computational resources.

3.2.1 Training Strategy and Hyperparameter Optimization

To ensure consistency in experimental conditions and comparability of results, a unified training approach was applied across all three networks:

- **Transfer Learning**:
  The models were initialized with weights pre-trained on the ImageNet dataset, enabling faster convergence by leveraging generalized visual features. The final fully connected layers were modified to match the number of output classes.
- **Optimization Algorithm**:
  The AdamW optimizer was employed *(learning rate = 3e-4, weight_decay = 1e-4)*, while the CosineAnnealingLR scheduler was used to control the learning rate during training.
- **Loss Function and Regularization**:
  The loss function used was CrossEntropyLoss with a label smoothing coefficient of 0.1. To mitigate the risk of overfitting, Dropout was applied, along with data augmentation techniques including RandomResizedCrop, HorizontalFlip, VerticalFlip, Rotation, ColorJitter, GaussianBlur, RandomAffine, and RandomErasing.

3.2.2 Training Dynamics and Curve Analysis

The empirical analysis of the training process indicates that all three architectures demonstrate rapid convergence during the initial training epochs**.**

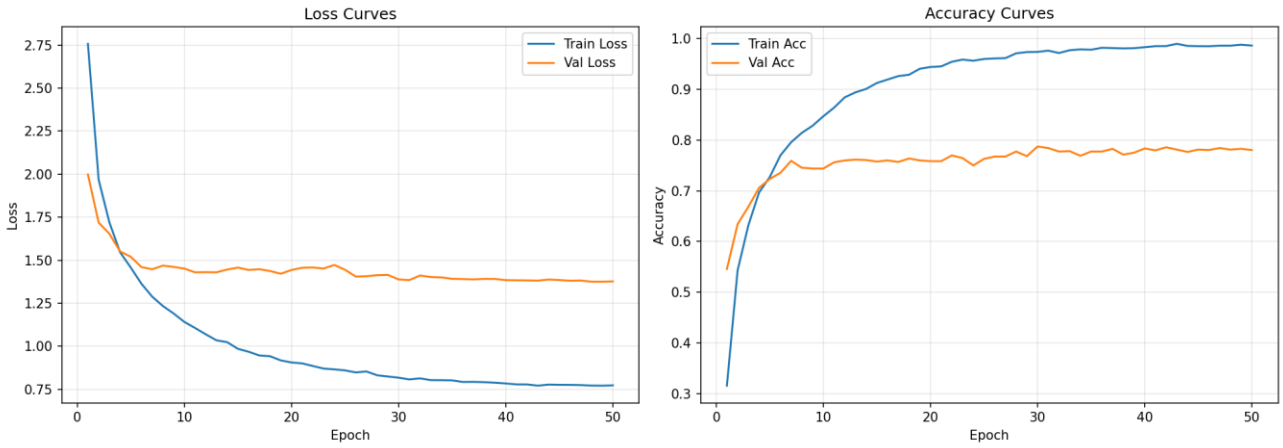

Figure 9. Training curves of the EfficientNet model (train/validation loss and accuracy)

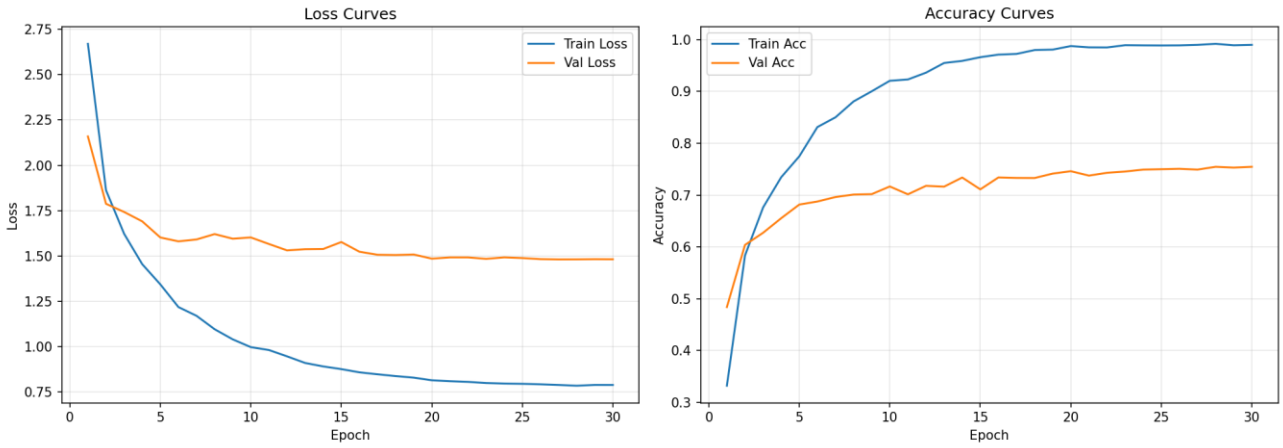

Figure 10. Training curves of the GoogLeNet model (train/validation loss and accuracy)

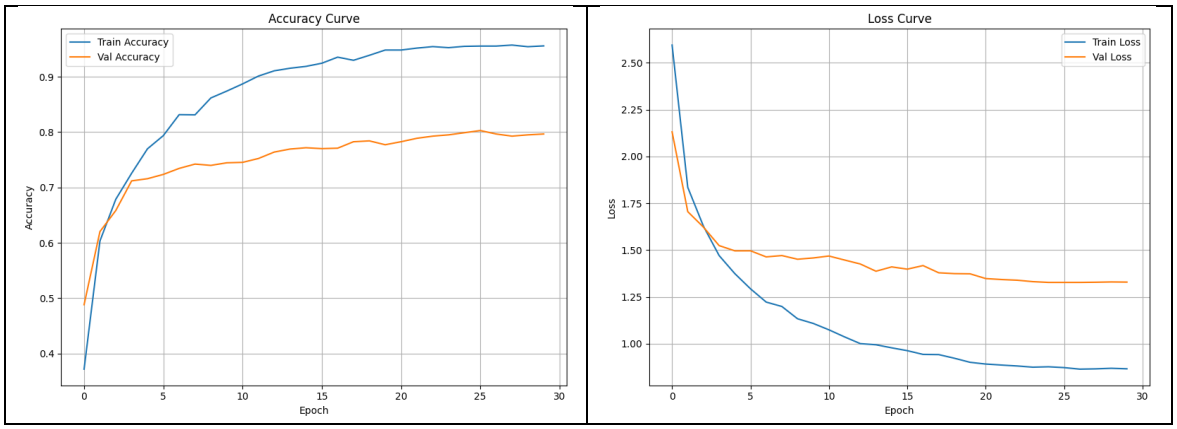

Figure 11. Training curves of the MobileNet model (train/validation loss and accuracy)

The analysis of the graphs (Figures 9, 10, 11) shows that the models achieve high accuracy on the training set (up to 95–99%). However, the stabilization of the validation curves indicates the presence of mild overfitting, which can be attributed to class imbalance in the dataset and high inter-class visual similarity.

3.2.3 Comparative Evaluation of Model Performance

The final evaluation of the models was conducted on the test set, with the results summarized in Table 2.

Table 2. Performance results of the models

Architecture	Train Accuracy	Validation Accuracy	Test Accuracy	Macro F1	Weighted F1
EfficientNet-B0	**98%**	79%	**78%**	**75%**	**78%**
MobileNetV3	95%	**80%**	77%	74%	76%
GoogleNet	99%	75%	73%	71%	73%

A comprehensive analysis of the obtained results allows the following conclusions to be drawn:

1. **Top-performing model**:
   On the test set, the best performance was achieved by the EfficientNet-B0 model, with the highest macro-F1 (0.759) and weighted-F1 (0.780) scores, along with an accuracy of 78.35%.
2. **Generalization capability**:
   MobileNetV3 demonstrated the highest validation accuracy (80.3%) while having the smallest number of parameters, making it the most hardware-efficient option.
3. **Impact of class imbalance**:
   All three models showed a noticeable drop in performance for underrepresented classes (e.g., *pepper_powdery_mildew* — 3 samples, *septoria_leaf_spot* — 15 samples), indicating a direct correlation with the class distribution in the dataset.
4. **Practical significance**:
   Due to its low computational complexity and relatively high accuracy, MobileNetV3 is considered the most suitable solution for real-time deployment on edge devices.

3.3 CLIP Model Architecture, Training Strategy, and Results Analysis

The CLIP (Contrastive Language–Image Pre-training) model was applied as a vision-language pre-trained architecture that provides expressive image representations through contrastive learning on large-scale image–text pairs. In this work, the *openai/clip-vit-base-patch32* variant was used as the backbone, where the vision encoder served as a feature extractor for the plant disease classification task.

3.3.1 Architectural Characteristics

Full fine-tuning of CLIP requires substantial computational resources, which was not feasible under the given conditions. Therefore, parameter-efficient fine-tuning (PEFT) was applied using the Low-Rank Adaptation (LoRA) method with the following configuration: *r = 2, lora_alpha = 8, lora_dropout = 0.10, target modules = {q_proj, k_proj, v_proj, out_proj}.* The backbone parameters were frozen, while only the LoRA adapters and the classification head

were made trainable. The total number of model parameters is 151,594,921, of which only 317,608 (approximately 0.21%) are trainable. This significantly reduces computational cost and helps mitigate the risk of overfitting.

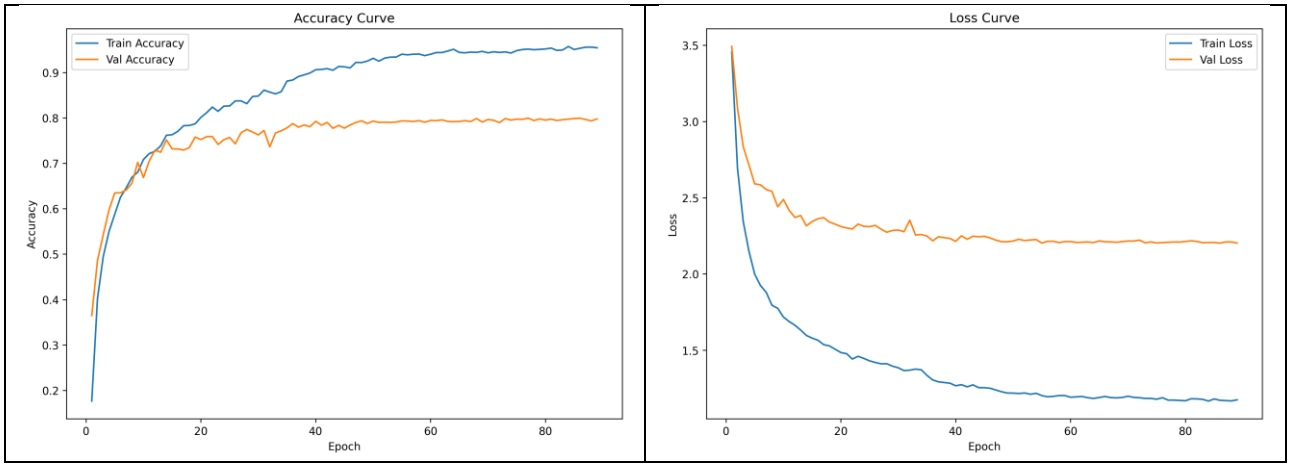

Figure 12. Training curves of the CLIP + LoRA model (train/validation loss and accuracy)

The training dynamics indicate stable convergence: both the train and validation loss curves decrease smoothly without sharp fluctuations, while the accuracy curves gradually increase and approach each other. The relatively small gap between train and validation results suggests a low level of overfitting compared to CNN-based models. On the test set, the model achieves 77.25% accuracy, with macro-F1 = 0.7709 and weighted-F1 = 0.7709. The performance of CLIP is comparable to EfficientNet-B0 and MobileNetV3, despite the fact that only 0.21% of the parameters are trainable, which highlights the strong generalization capability of vision-language pretraining. A detailed analysis of the classification results shows that the model performs particularly well on classes such as *leaf_curl*, *black_leaf_streak*, and *pepper_blossom_end_rot* (F1 = 1.00), while a decrease in performance is observed for visually similar classes and those with limited representation (e.g., *bacterial_leaf_streak*).

3.4 DINOv2 Model Architecture, Training Strategy, and Results Analysis

The DINOv2 model was applied as an advanced transformer-based approach for solving the plant disease classification task. It is a self-supervised Vision Transformer that learns high-quality feature representations from large-scale data without requiring fully labeled datasets. In this work, the DINOv2-large backbone was used as a feature extractor, followed by fine-tuning. Most of the model parameters were frozen, and training was performed using the Low-Rank Adaptation (LoRA) method, which enables efficient adaptation of large models without full fine-tuning. Additionally, a custom classification head was applied, which combines the CLS token and the average of patch embeddings, providing a more expressive feature representation.

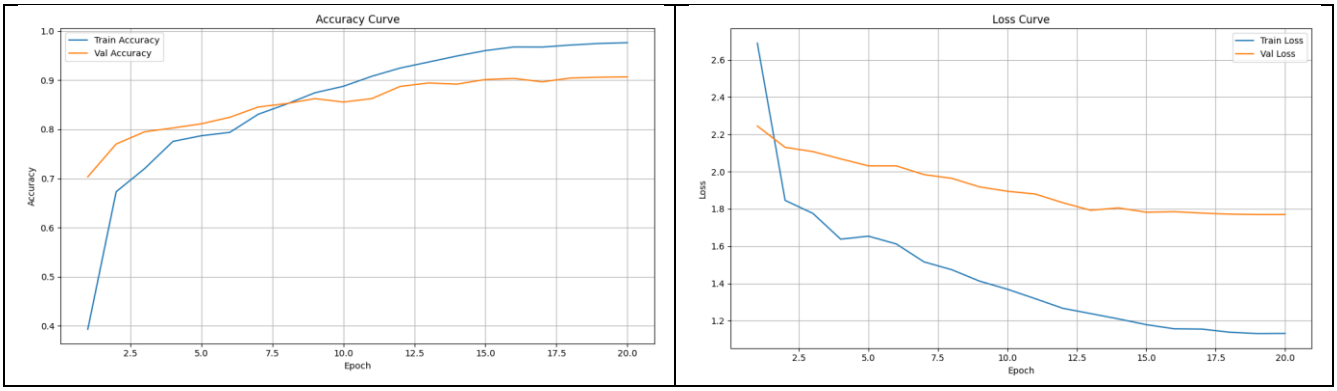

Figure 13. Training curves of the DINOv2 model (train/validation loss and accuracy)

The analysis of the training curves shows that the model achieves stable and efficient convergence. The train loss continuously decreases, while the validation loss also gradually declines without sharp fluctuations, indicating stability in the training process. Observation of the accuracy curves shows that the validation accuracy reaches high values (~90%+) and approaches the train accuracy, demonstrating strong generalization capability. The relatively small gap between train and validation results further indicates a low level of overfitting. The final evaluation of the model was conducted on the test set, yielding approximately 90.2% accuracy. This significantly outperforms the applied convolutional models and highlights the high effectiveness of the DINOv2 model for this task. A detailed analysis of the classification results shows that the model achieves high precision and recall for most major classes, such as *canker*, *powdery_mildew*, *rust*, *leaf_curl*, and *head_scab*, where the scores approach or reach 1.0. At the same time, a performance drop is observed for some rare classes, such as *septoria_leaf_spot* and *pepper_powdery_mildew*, which is due to the limited number of samples and class imbalance.

3.5 Comparative Analysis of All Models

Table 3. Final comparative evaluation of all five models

	Model	Types	Total Parameters	Trainable Parameters	Test Acc	Macro-F1	Weighted-F1
1	GoogLeNet	CNN	5.6 M	5.6 M	0.7383	0.7141	0.7361
2	MobileNetV3-Large	CNN	4.3 M	4.3 M	0.7726	0.7457	0.7693
3	CLIP ViT-B/32 + LoRA	Transformer (V-L)	151.6 M	0.32 M	0.7725	0.7709	0.7709
4	EfficientNet-B0	CNN	4.1 M	4.1 M	0.7835	0.7592	0.7803
5	DINOv2-Large + LoRA	SSL Transformer	**304 M**	**~6 M**	**0.9019**	**0.88**	**0.90**

Conclusion

Based on the conducted experiments, it can be concluded that the best performance on the final cleaned and preprocessed dataset was achieved by the DINOv2 model, providing the highest accuracy and strong generalization capability. This is largely due to its transformer-based architecture and its ability to effectively capture global features. The convolutional neural networks—EfficientNet, MobileNet, and GoogLeNet—produced relatively comparable results; however, signs of overfitting were observed during their training. This issue is mainly caused by class imbalance in the dataset, as well as the limited representation of certain classes. At the same time, the DINOv2 model was able to partially mitigate this problem, delivering more stable and balanced performance across different classes. Overall, the experiments demonstrate that transformer-based approaches, particularly self-supervised models, offer a more effective solution for complex image classification tasks, especially in scenarios with imbalanced data

References

[1] *EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks*. ICML. https://arxiv.org/pdf/1905.11946

[2] *Going deeper with convolutions* https://arxiv.org/pdf/1409.4842

[3] *Searching for MobileNetV3 https://arxiv.org/pdf/1905.02244*

[4] *DINOv2: Learning Robust Visual Features without Supervision* https://arxiv.org/pdf/2304.07193

[5] *LORA: LOW-RANK ADAPTATION OF LARGE LANGUAGE MODELS https://arxiv.org/pdf/2106.09685*

[6] Contrastive Localized Language-Image Pre-Training https://arxiv.org/pdf/2410.02746